

Dedekind Finiteness Is Not Open In The Free-Monoid Quotient Coding

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Source And Question

The source paper is Tomasz Kania, *Polish spaces for countable and separable structures through quotient encodings*, arXiv:2604.15843. In Section 12, under “Dedekind finiteness for Banach algebras”, the paper asks whether the closed set of Dedekind finite Banach-algebra quotients is non-open in the Wijsman topology, or whether natural subclasses exhibit sharper completeness phenomena.

This packet answers the non-openness part for the standard unital Banach-algebra quotient generator used in the paper:

$$A = \ell_1(S),$$

where S is the free monoid on countably many generators.

Definitions

A unital Banach algebra B is Dedekind finite, or directly finite, if

$$ab = 1 \implies ba = 1$$

for all $a, b \in B$. Otherwise B is Dedekind infinite.

For a closed ideal $K \subset A$, the Wijsman topology is the topology of pointwise convergence of quotient norms:

$$K_i \rightarrow K \iff \text{dist}(x, K_i) = \|x + K_i\|_{A/K_i} \rightarrow \|x + K\|_{A/K} \quad (x \in A).$$

Proof Intuition

Dedekind infiniteness can be hidden arbitrarily far out in the free generating sequence. Put a one-sided inverse pair on the generators s_{2n-1}, s_{2n} and send all other generators to zero. Each quotient is Dedekind infinite. But every fixed finite word only uses finitely many generators, so for large n the quotient cannot see the one-sided inverse pair on that finite part; there it agrees with the scalar quotient. Finite-support approximation then upgrades this agreement to Wijsman convergence on all of $\ell_1(S)$.

Theorem 1. *Let S be the free monoid on generators $s_1, s_2, \dots$, and let $A = \ell_1(S)$ be the standard unital free Banach-algebra quotient generator. In $\text{Ideal}(A)$ with the Wijsman topology, the class of Dedekind finite proper quotients is not open.*

Proof. Let $1 \in S$ denote the empty word. Define the augmentation character $\chi : A \rightarrow \mathbb{C}$ by

$$\chi(1) = 1, \quad \chi(w) = 0 \quad (w \neq 1),$$

extended linearly and continuously. This is a contractive unital algebra homomorphism, and $K_\infty := \ker \chi$ gives the scalar quotient $A/K_\infty \cong \mathbb{C}$, which is Dedekind finite.

Let $H = \ell_2(\mathbb{N}_0)$. Let $Re_j = e_{j+1}$ be the forward unilateral shift and let $Le_0 = 0$, $Le_{j+1} = e_j$ be the backward shift. Then

$$LR = I, \quad RL = I - p_0,$$

where p_0 is the rank-one projection onto $\mathbb{C}e_0$.

For each $n \geq 1$, define a contractive unital homomorphism $\pi_n : A \rightarrow B(H)$ by assigning

$$\pi_n(s_{2n-1}) = L, \quad \pi_n(s_{2n}) = R, \quad \pi_n(s_j) = 0 \quad (j \notin \{2n-1, 2n\}),$$

and extending by the universal ℓ_1 -free-monoid construction. Put $K_n = \ker \pi_n$. In A/K_n , the cosets of s_{2n-1} and s_{2n} satisfy

$$(s_{2n-1} + K_n)(s_{2n} + K_n) = 1 + K_n,$$

but

$$(s_{2n} + K_n)(s_{2n-1} + K_n) \neq 1 + K_n,$$

because $\pi_n(s_{2n}s_{2n-1} - 1) = RL - I = -p_0 \neq 0$. Thus every quotient A/K_n is Dedekind infinite.

It remains to prove $K_n \rightarrow K_\infty$ in the Wijsman topology. First take an element $y \in A$ with finite support, and suppose all non-empty words in its support use only generators among $s_1, \dots, s_m$. If $2n-1 > m$, then $\pi_n(w) = 0$ for every non-empty word w in the support of y , whence

$$\pi_n(y) = \chi(y)I.$$

Since $y - \chi(y)1 \in K_n$, we get

$$\text{dist}(y, K_n) \leq |\chi(y)|.$$

Conversely, π_n is contractive and vanishes on K_n , so for every $k \in K_n$,

$$|\chi(y)| = \|\pi_n(y)\| = \|\pi_n(y - k)\| \leq \|y - k\|.$$

Hence $\text{dist}(y, K_n) = |\chi(y)|$ for all sufficiently large n . The same identity holds for K_∞ : $\text{dist}(y, K_\infty) = |\chi(y)|$, because χ is a norm-one quotient map onto $\mathbb{C}$.

Now let $x \in A$ be arbitrary and choose finite-support y with $\|x - y\| < \varepsilon$. Distance to a closed set is 1-Lipschitz in the point, so for all sufficiently large n ,

$$\begin{aligned} |\text{dist}(x, K_n) - \text{dist}(x, K_\infty)| &\leq |\text{dist}(x, K_n) - \text{dist}(y, K_n)| \\ &\quad + |\text{dist}(y, K_n) - \text{dist}(y, K_\infty)| + |\text{dist}(y, K_\infty) - \text{dist}(x, K_\infty)| \\ &\leq 2\varepsilon. \end{aligned}$$

As $\varepsilon > 0$ is arbitrary, $\text{dist}(x, K_n) \rightarrow \text{dist}(x, K_\infty)$ for every $x \in A$. Thus $K_n \rightarrow K_\infty$ Wijsman.

We have exhibited a Dedekind finite quotient A/K_∞ that is a Wijsman limit of Dedekind infinite quotients A/K_n . Therefore the Dedekind finite class is not open. $\square$

Verification Notes

The proof uses only the standard free-monoid generator from the source paper and the elementary unilateral-shift one-sided inverse $LR = I$, $RL \neq I$. The Wijsman convergence proof is pointwise on quotient norms, first for finite-support elements and then by density of finitely supported elements in $\ell_1(S)$.

Limitations And Novelty Check

Cheap run indexes were searched for arXiv:2604.15843, Dedekind finiteness, Banach-algebra quotient coding, and related stable-rank/operator-algebra keywords. No prior packet for this source question was found.

This is a partial result for the paper’s list of open questions. It answers the non-openness question for the standard unital Banach-algebra quotient generator $A = \ell_1(S)$. It does not prove an optimal completeness rank, and it does not address every possible quotient generator or every admissible hyperspace topology.

References

- T. Kania, *Polish spaces for countable and separable structures through quotient encodings*, arXiv:2604.15843.

Human Review Recommendation

Check that the source’s intended “present coding” for Banach algebras includes the standard unital free-monoid generator $A = \ell_1(S)$, and verify the quotient-norm convergence calculation. If accepted, this should be recorded as a scoped partial answer to the Dedekind-finiteness item in Section 12.